👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

Understand Equations and Their Solutions

Lesson 8-1 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

COLD CASE MISSION

The Equation Vault

You are Detective Ruiz's new code analyst. A tip just came in: 'A number of stolen gems plus 8 more equals 20 total.' The evidence vault only opens when the clue is translated into a correct equation. Master writing equations and the case cracks wide open.

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write an equation for a real situation and check whether a given value is a solution by substituting it.

Language Objective: I can explain a solution using the words substitute, solution, true, and equal sign.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Equation · **Variable** · **Equal sign** · **Expression** · **Constant** · **Unknown** · **Solution** · **Substitute** · **Estimate**

Level 3: try the blanks from memory first, then check the bank.

1. A math sentence that says two amounts are equal, with an = sign, is an _____.

2. A letter that stands for an unknown number is a _____.

3. The symbol = that shows two amounts are the same is the _____.

4. A group of numbers and variables with no equal sign is an _____.

5. A fixed number that does not change is a _____.

6. The value you are trying to find in an equation is the _____.

7. A value that makes the equation true when _____ for the variable is its solution.

8. Replace the variable with a number to _____ whether it is a solution.

9. A close first guess that helps you start is an _____.

Write About the Math

How many paintings has the detective recovered? How did writing an equation help you understand the situation?
¿Cuántas pinturas ha recuperado la detective? ¿Cómo te ayudó escribir una ecuación a entender la situación?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Equation** — *Ecuación*
- ☐ **Variable** — *Variable*
- ☐ **Equal sign** — *Signo igual*
- ☐ **Expression** — *Expresión*
- ☐ **Constant** — *Constante*

Model: *I let a variable stand for the unknown number of gems.* — Students choose a variable (such as n) for the unknown gems, build n + 8 = 20, and connect 'plus 8 more' to + 8 and 'equals 20 total' to = 20.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The detective recovered ___ paintings because p + 14 = 37 means p = 37 − ___ = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.

- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Variable
3. **Notes 3:** Equal sign
4. **Notes 4:** Expression
5. **Notes 5:** Constant
6. **Notes 6:** Unknown
7. **Notes 7:** Solution
8. **Notes 8:** Substitute
9. **Notes 9:** Estimate
10. **Watch for this mistake:** *A common mistake in Write Equations is writing subtraction and division clues in the wrong order — putting the number from the clue first instead of the variable first. For 'a number decreased by 8 equals 5,' students often write 8 − n = 5 instead of the correct n − 8 = 5; for 'a number divided by 4 equals 6,' they write 4 ÷ n (or 4n) instead of n / 4 = 6. Always start with the variable, then attach the operation in the order the words describe it, and check by substituting your answer back in.*
11. **Try It 1:** A. 5b = 35 — *'5 times the number' means 5 times b, so 5b = 35. Check: 5 × 7 = 35.*
12. **Try It 2:** A. m − 9 = 24 — *'9 fewer' means subtract 9, so m − 9 = 24. Check: 33 − 9 = 24.*